I built 3 Power BI dashboards. Here's what the data revealed.

8 dashboard tabs · 10 business questions answered · 3 stakeholder audiences

PostgreSQL · dbt · Airflow · Power BI

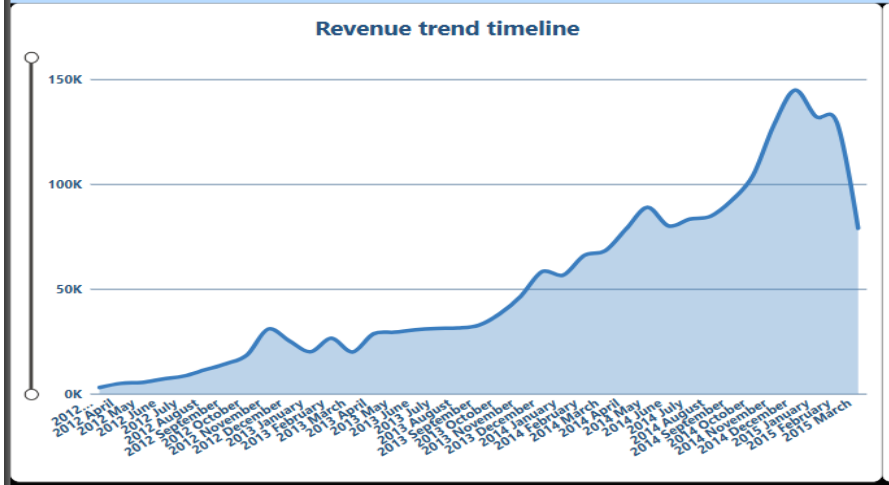
M E C P

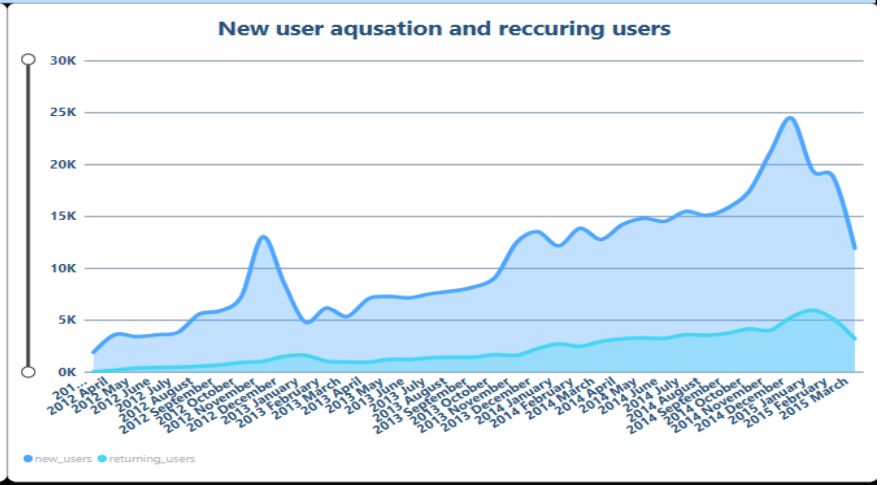
Maven
E-Commerce
Portfolio
Project

EXECUTIVE DASHBOARD

Revenue grew 174% YoY

Total Revenue	Total Orders	AOV	Total Conversion rate	YoY Revenue Growth
1.94M	32.313K	59.99	6.8%	173.5%





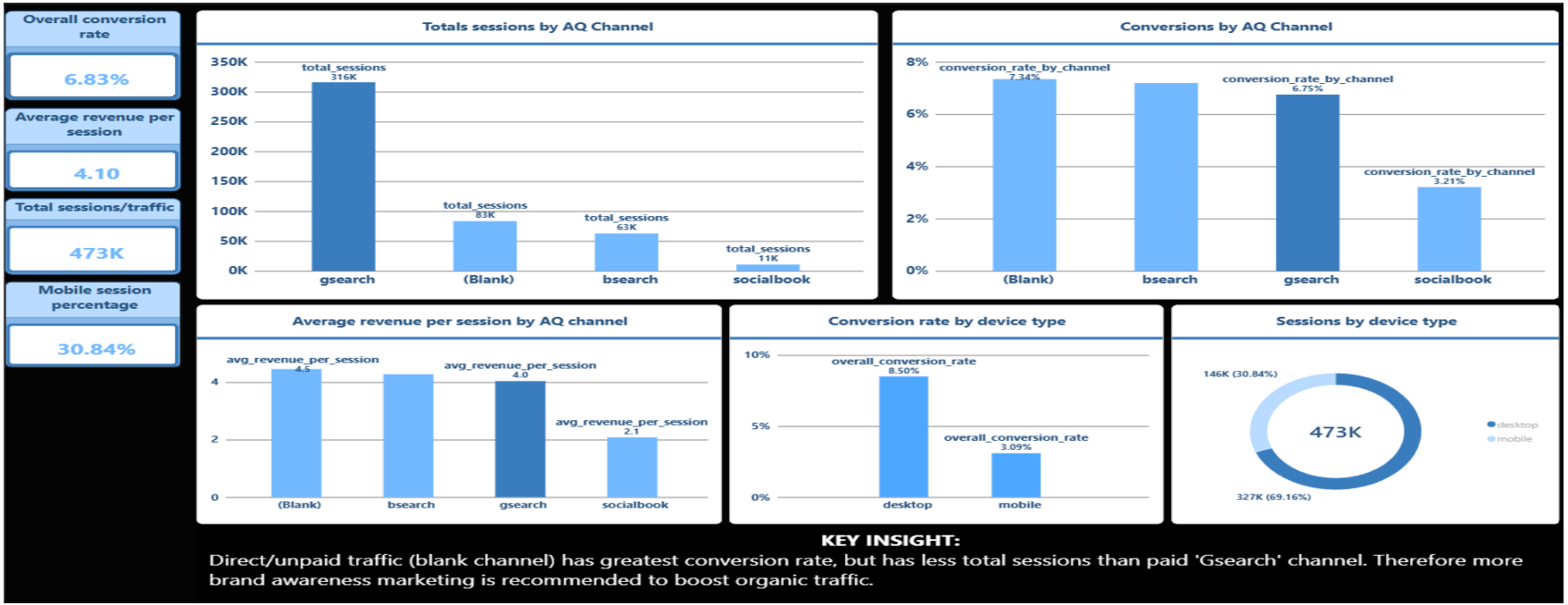
KEY INSIGHT:

- Revenue grew 174% YoY (2013→2014), driven by new product launches and channel expansion.
- Overall conversion rate is 6.83%, with desktop users converting 2.75x better than mobile (8.5% vs 3.09%).
- Direct/organic traffic shows highest conversion quality despite lower volume than paid search.

Total Revenue: $1.94M · Orders: 32K · AOV: $59.99 · Conversion Rate: 6.8%

ACQUISITION & CHANNELS

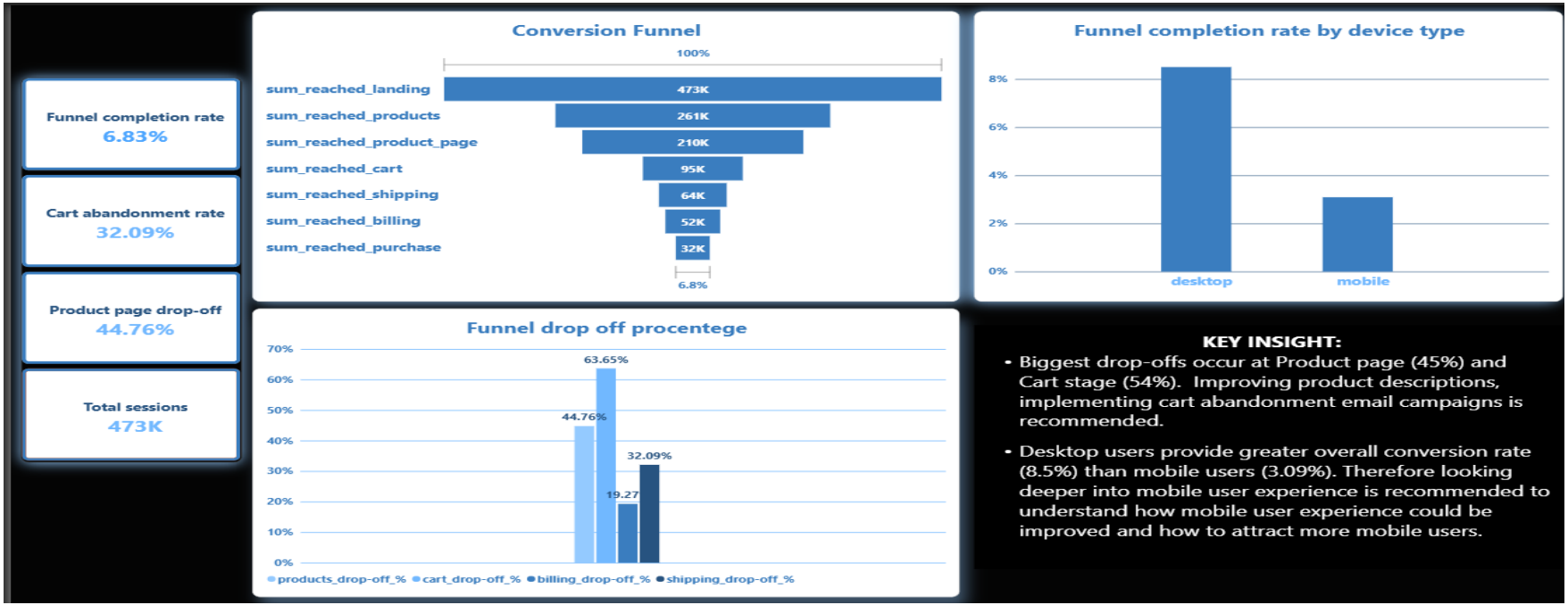
Direct traffic converts 2x better than paid



Direct/organic: 7.34% conv rate · gsearch: 6.75% · socialbook: 3.21%
Desktop: 8.5% conv rate · Mobile: 3.09% — 2.75x gap

CONVERSION FUNNEL

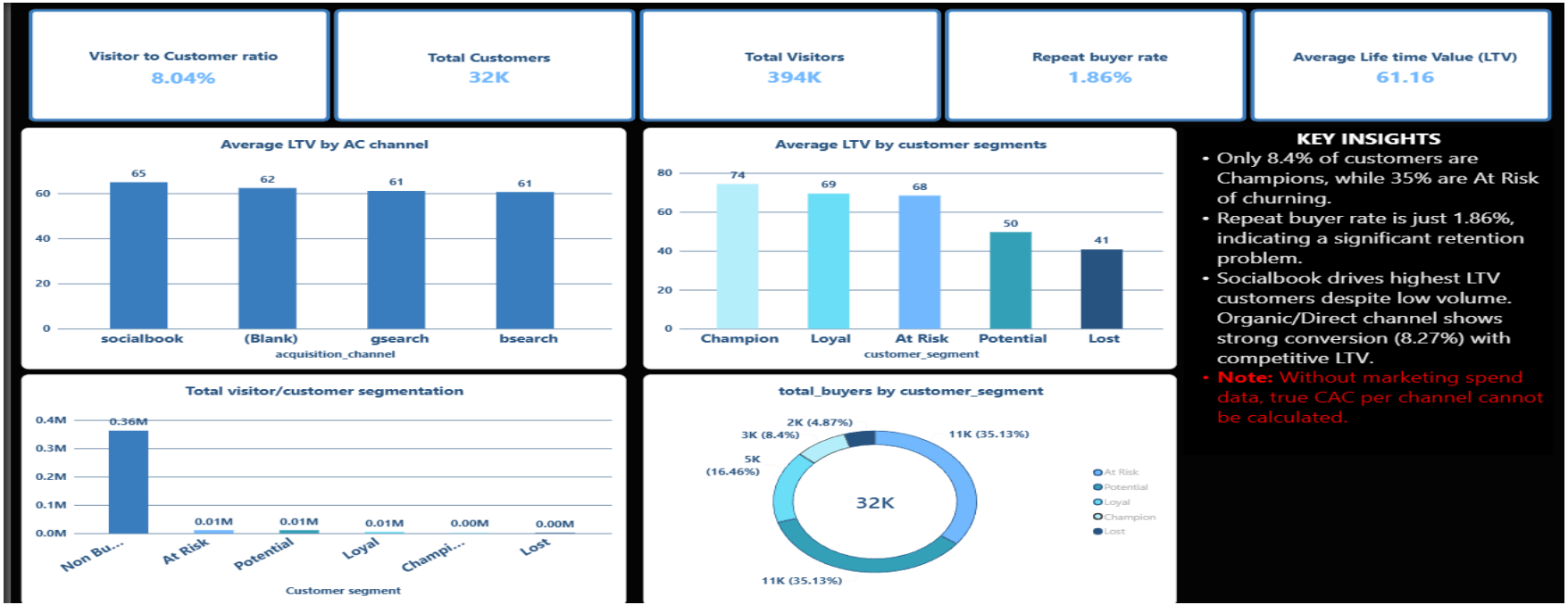
93% of sessions never convert



Biggest drop-offs: Product page (44.76%) · Cart stage (63.65%)
Recommendation: Improve product descriptions + cart abandonment campaigns

CUSTOMER SEGMENTATION & LTV

35% of customers are At Risk of churning



Champions: 8.4% · At Risk: 35.13% · Repeat buyer rate: 1.86%
Avg LTV: $61.16 · Socialbook drives highest LTV despite lowest volume

CUSTOMER RETENTION

Most cohorts retain <20% by Month 2

Average cohort size	995.18
Average Month 1 retention	0.84
repeat_buyer_rate	1.86%
Best retention Cohort	2014-04

KEY INSIGHT:

- Customer retention drops sharply after the first purchase — most cohorts show less than 20% retention by Month 2.
- Combined with a 1.86% repeat buyer rate, this confirms a significant retention problem. Recommend implementing post-purchase email sequences, loyalty programs, and personalized offers to improve Month 1–2 retention.

RETENTION MATRIX

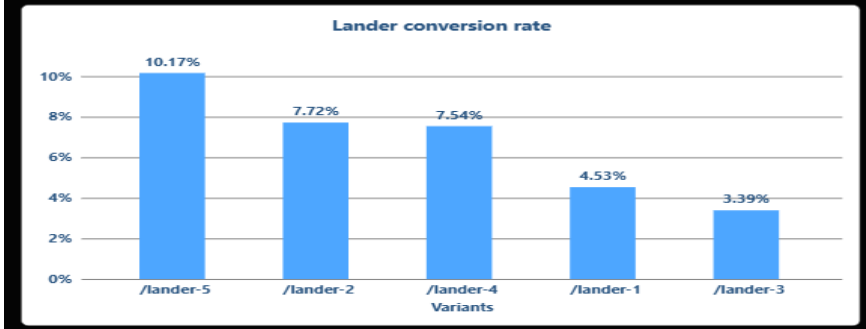
cohort_label	1	2	3	4	Total
2012-06	72.00%				72.00%
2012-08	44.00%				44.00%
2012-09	35.00%	105.00%			70.00%
2012-10	54.00%				54.00%
2012-11	33.00%	16.00%		16.00%	21.67%
2012-12	20.00%	60.00%			40.00%
2013-01	103.00%		26.00%		64.50%
2013-02	41.00%	82.00%			61.50%
2013-03	79.00%	52.00%			65.50%
2013-04	18.00%	37.00%			27.50%
2013-05	89.00%	53.00%	18.00%		53.33%
2013-06	68.00%	51.00%			59.50%
2013-07	84.00%	84.00%	17.00%		61.67%
2013-08	100.00%	67.00%	17.00%		61.33%
2013-09	130.00%	32.00%	16.00%		59.33%
2013-10	86.00%	14.00%	29.00%		43.00%
2013-11	59.00%	59.00%	24.00%		47.33%
2013-12	77.00%	58.00%	10.00%		48.33%
2014-01	104.00%	62.00%	42.00%		69.33%
2014-02	80.00%	90.00%	20.00%		63.33%
2014-03	105.00%	38.00%	19.00%		54.00%
2014-04	181.00%	41.00%			111.00%
2014-05	75.00%	52.00%	22.00%		49.67%
2014-06	90.00%	41.00%	16.00%		49.00%
2014-07	56.00%	95.00%	8.00%		53.00%
2014-08	123.00%	54.00%	8.00%		61.67%
2014-09	108.00%	58.00%	29.00%	7.00%	50.50%
2014-10	114.00%	63.00%	19.00%		65.33%
2014-11	118.00%	41.00%	26.00%		61.67%
Total	83.81%	57.04%	20.21%	11.50%	57.86%

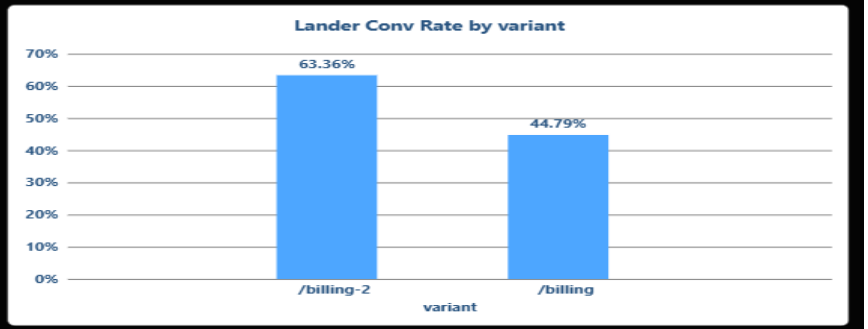
Avg Month 1 retention: 0.84% · Best cohort: 2014-04
Recommendation: Post-purchase email sequences + loyalty programs

A/B TESTS

/lander-5 converts 3x better than /lander-3

Best Lander Variant	Best Billing Variant	Total Billing Sessions	Total Lander Sessions
/lander-5	/billing-2	52K	335K



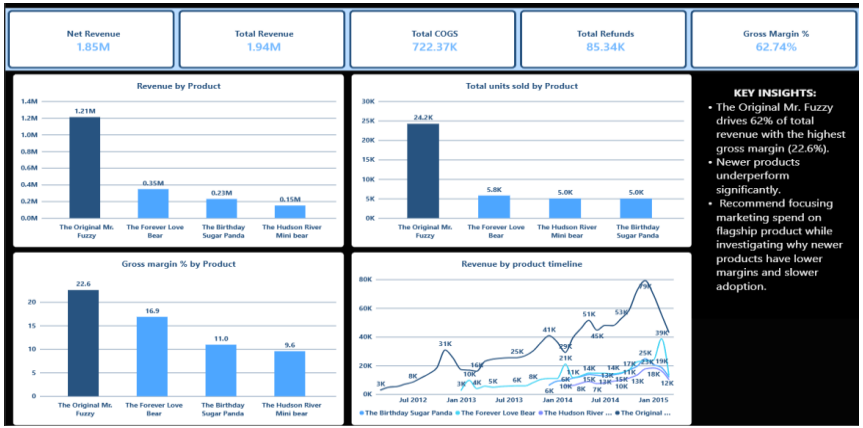


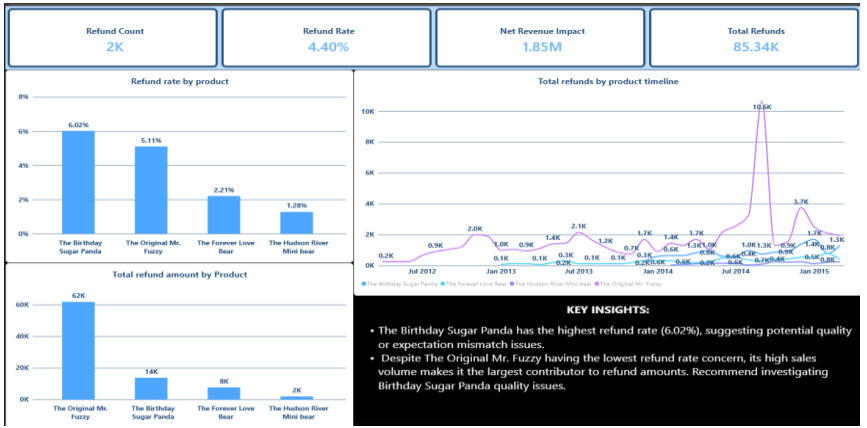
Key Insight:

- A/B tests show clear winners: /lander-5 converts 3x better than /lander-3 (10.17% vs 3.39%). The redesigned billing page (/billing-2) improved checkout completion by 18.6 percentage points (63.36% vs 44.79%).
- Recommend permanently implementing /lander-5 and /billing-2 across all traffic.
- NOTE: Statistical significance of these results has not been validated. Sample sizes and p-values should be calculated before making final implementation decisions. Results suggest /lander-5 and /billing-2 as strong candidates for permanent implementation pending statistical validation.

Lander winner: /lander-5 (10.17%) · Loser: /lander-3 (3.39%)
Billing winner: /billing-2 (63.36%) · vs /billing (44.79%) — +18.6pp improvement

One product drives 62% of total revenue





$1.94M	62.74%	4.40%	$85.34K
Total Revenue	Gross Margin	Refund Rate	Total Refunds

Mr. Fuzzy: $1.21M revenue · 22.6% margin · Birthday Sugar Panda: highest refund rate (6.02%)
Recommend: Focus marketing on flagship · Investigate Sugar Panda quality issues

3 dashboards.
10 business questions.
1 complete pipeline.

Full project on GitHub:

github.com/BenatrixB/MECP_Ecommerce_data_portfolio_project

Questions or feedback? Drop them in the comments.